Handling Data 1 — Types of data · Presenting data · Averages

Calculator allowed. Squared paper and a protractor for the drawing questions. Marks in brackets. ★ = harder question.

1 Say whether each type of data is discrete, continuous or categorical: **a** the number of goals scored in a match, **b** the height of a plant, **c** a person's favourite colour, **d** shoe size. [2]

2 A pie chart shows how the 48 students in a year group travel to school. The sector angles are: bus $120°$, walk $135°$, car $60°$, bicycle $45°$. [3]

a How many students come by bus?

b Next year 9 students cycle, out of the same 48. What should the bicycle sector's angle be?

3 All 180 students in Year 11 were asked whether they walk or cycle to school. 125 said they walk, 50 said they cycle, and 15 do neither. How many both walk and cycle? [2]

4 The table shows the medals won by a team in two years. [5]

	Gold	Silver	Bronze
Year 1	12	18	15
Year 2	26	17	20

a Draw a comparative (dual) bar chart to show this data.

b Comment on the difference between the two years.

5 A café serves a set menu: soup or salad to start, then fish or vegetable curry. One lunchtime there were 60 customers. Copy and complete the two-way table. [3]

	Fish	Vegetable	Total
Soup	18		27
Salad		8	
Total			60

6 A drinks company's bar chart claims cola sells "more than twice as much" as other drinks. The vertical axis starts at 300 (not 0), and the cola bar is drawn wider than the others. Give two reasons why the chart is misleading. [2]

7 Find the mean, mode and median of $3,\ 8,\ 5,\ 12,\ 8,\ 0,\ 6,\ 9,\ 3$. [3]

8 To get a distinction, Eleni's mean mark over five tests must be at least 85. Her first four marks are 86, 94, 75 and 88. What is the lowest mark she can get in the fifth test? [3]

9 In a town, the mean number of parcels delivered per day was 2 in March and 1.6 in April. What was the mean number per day over the two months together? Give your answer to 3 s.f. [4]

Total: 27 marks

Check: pie chart angle = fraction × 360° · a two-way table's rows and columns must each add to their totals · mean needs the total and the count — never average averages.